Evaluating the potential of LLMs as semantic task annotators

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Introduction: Large language models (LLMs) have the potential to accelerate critical, but often time consuming research tasks including data annotation, stimulus labeling and feature norming. Recent work shows that using LLMs for semantic feature norming can improve data quality and predict human similarity judgements more accurately than existing human-derived datasets [1], [2]. But prior studies also demonstrate that model-human correspondence is highly variable across many domains of world knowledge [3], raising a key question: to what extent can LLMs reproduce human-like judgments in semantic annotation tasks? In this work, we evaluate how closely LLM responses align with human judgements across three tasks: (a) sentence-level semantic labeling, (b) yes-no question answering and (c) sentence plausibility judgments.

Methods: (a) For sentence-level semantic labeling, three researchers rated whether a sentence contained information about locations and emotions. 298 sentences were taken from two stories from *The Moth* podcast. (b) For yes-no question answering, 292 online participants answered two conceptual questions ("Did this exist 1000 years ago?" and "Does this make a characteristic sound?") for 50 concrete objects from the THINGS dataset [4] presented as words. (c) For plausibility judgements, we used two sentence sets from studies [5] (658 participants, ~23 ratings/sentence; referred to as Plausibility dataset 1 henceforth) and [6] (96 participants, 48 ratings/sentence; referred to as Plausibility dataset 2 henceforth) in which participants rated sentence plausibility on a 1–7 Likert scale. All datasets are in English. We compared the output of three open source LLMs of varying sizes–Llama 8b, Phi 14b and Llama 70b–against average human annotator ratings for each task. Inter-rater agreement is quantified using a leave-one-out approach, where for each iteration, a human participant is held out and the mean response of the remaining participants serves as the reference to compute the mean squared error (MSE). In addition to evaluating the models separately, we also consider a "wisdom of crowds" (WOC) approach, where we obtain the median response from a group of models. We also compute the Krippendorff's alpha for all human raters and all LLM raters for each dataset as another metric to quantify inter-annotator agreement.

Results: We find that, for all datasets, human-model MSE scores are comparable to (e.g., semantic sentence labeling - emotions, one-sample t-test, FDR corrected, $\mathbf{p>0.05}$) or lower (e.g., Plausibility dataset 1; $\mathbf{p<0.001}$) than human-human MSE scores, suggesting that LLMs are good approximators of average human responses to semantic tasks (Figure 1). However, Levene's test revealed substantial variability in human-model agreement at the individual item-level relative to human-human agreement ($p<0.05$ in 4/6 datasets). For instance, in yes-no question answering tasks, the 95th percentile of human MSE scores is 0.23 but 0.69 for Llama 8b. Using a WOC approach does not ameliorate this discrepancy between humans and models (MSE(95%)=0.56), possibly because LLMs generally agree with one another across all tasks. Krippendorff's alpha reveals moderate human-human agreement but higher model-model agreement across most tasks (Table1).

Conclusion: LLMs demonstrate strong potential for automating semantic annotation tasks, achieving statistically similar (sometimes lower) error rates compared to average human judgments. But, models continue to make unnatural mistakes; verifying model responses may be a critical step when using LLMs to approximate human-like behaviors.

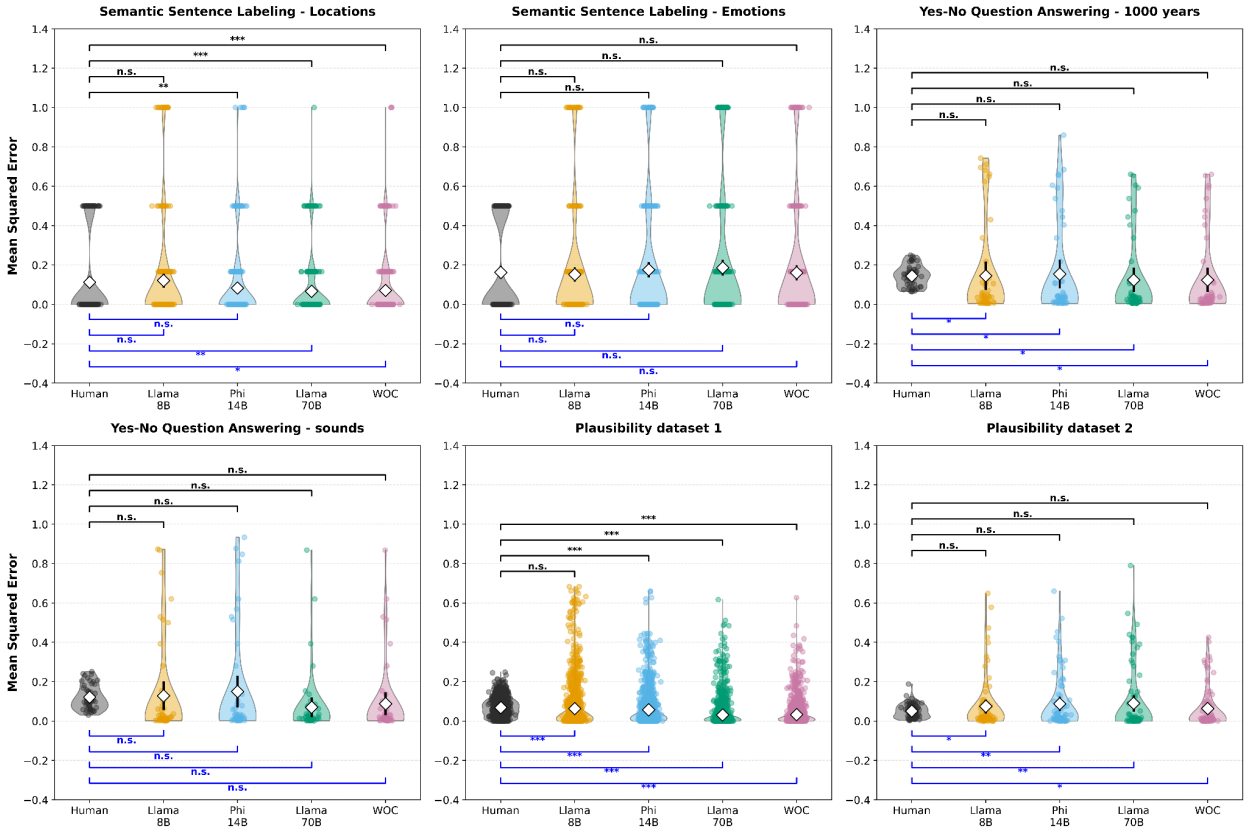

Figure 1: MSE distribution of the human-human, human-model and human-WOC model across sentences/words in each task. White circles correspond to mean MSE scores and error bars indicate 95% confidence interval of the mean. Black brackets indicate significant differences in mean error between models and the human baseline (paired t-test; FDR corrected; * p<0.05, ** p<0.01, *** p<0.001, n.s.: not significant). Blue brackets indicate significant differences in error variance (Levene's test; significance levels mirrored). While models' mean MSE scores are comparable to or lower than human baselines, they consistently exhibit significantly higher variance, suggesting that model errors are more prone to outliers than humans.

Tasks	Human-human Krippendorff's α	LLM-LLM Krippendorff's α
Semantic Sentence labeling (locations)	0.59	0.69
Semantic Sentence labeling (emotions)	0.43	0.68
Yes-No question answering (1000 years)	0.41	0.52
Yes-No question answering (sounds)	0.5	0.71
Plausibility Dataset 1	0.57	0.59
Plausibility Dataset 2	0.72	0.69

Table 1: Comparison of Human-Human and LLM-LLM Krippendorff's α across all tasks. We observe that LLM-LLM agreement is consistently comparable to or higher than Human-Human agreement, indicating that different LLMs tend to make similar judgments. This strong alignment might limit the effectiveness of the WOC approach.

References: [1] K. Mukherjee et al., 2023, arXiv [2] S. Suresh et al., 2025, arXiv [3] A. A. Ivanova et al., 2025, Trans. Assoc. Comput. Linguist. [4] M. N. Hebart et al., 2023, eLife [5] E. Fedorenko et al., 2020, Cognition [6] A. A. Ivanova et al., 2021, Neurobiol. Lang.